Operations on Data Structures

Q. Discuss the various fundamental operations that can be performed on data structures, explaining each with a suitable example.

> **Definition to Remember**

> **Op
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the eff

The Six Fundamental Operations

| Operation | Definition | Example |

| :--- | :

Traversal Algorithm Example (Array)

```text

Algorit

## Trade-offs Between Data Structures

| DS | Fast Operations | Slow Operations |

| :--- | :

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> **Must-Write Points (for 10 marks)**

> 1. Si
```

> **Quick Recall**

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structu

Algorithmic Implementations & Executable C Codes

1. Array Operations: Traversal, Insertion & Deletion (C Code)

```c
#includ

void traverse(int arr[], int n) {
printf("

int insertElement(int arr[], int n, int capacity, int element, int pos) {
if (n >=

int deleteElement(int arr[], int n, int pos) {
if (pos

int main() {
int arr[

printf("--- Initial Array ---\n");
travers
```

printf("\n--- Inserting 25 at Index 2 ---\n");

n = ins

printf("\n--- Deleting Element at Index 3 (30) ---\n");

n = de

return 0;

}

---

2. Searching Algorithms: Linear Search vs. Binary Search (C Code)

```c

#includ

// Linear Search - O(n)

int line

// Binary Search - O(log n) [Requires Sorted Array]

int bina

int main() {

int arr[

int linRes = linearSearch(arr, n, target);

printf("

int binRes = binarySearch(arr, n, target);

printf("

return 0;

}

---
```

3. Sorting & Merging Operations (C Code)

```c
#includ

// Bubble Sort Operation - O(n^2)
void bu

// Merging Two Sorted Arrays - O(m + n)
void m

int main() {
int A[]

mergeArrays(A, m, B, n, C);
printf("

return 0;
}
```